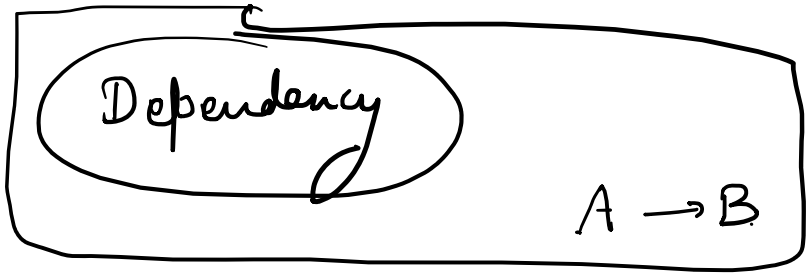
3NF



1) A non-key attribute shouldn't help determine an another attribute.

student-enrollment

st_id	std_name	c_id	c_name	inst_id	inst_name
1	ABC	t01	SWE	301	pqr
2	XYZ	t02	AI	309	uvw

Composite PK

What? ⇒ unrelated entity (cause)

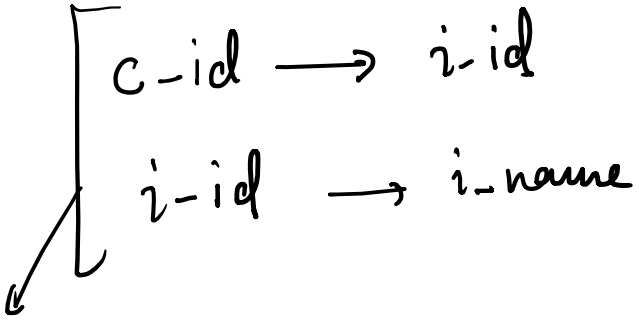
Sol^{n} ⇒ move it out to a separate table.

Redundancies

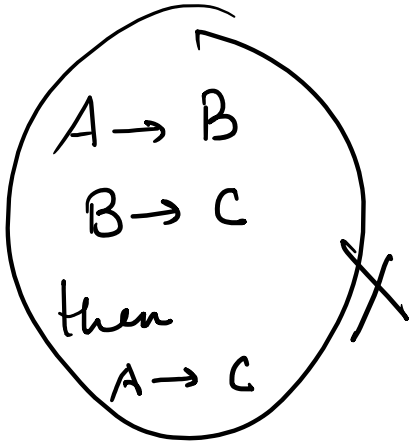
Why → Redundancies

(Why to follow 3NF)

↳ Transitive dependency shouldn't exist in the same table



c_id → i_name

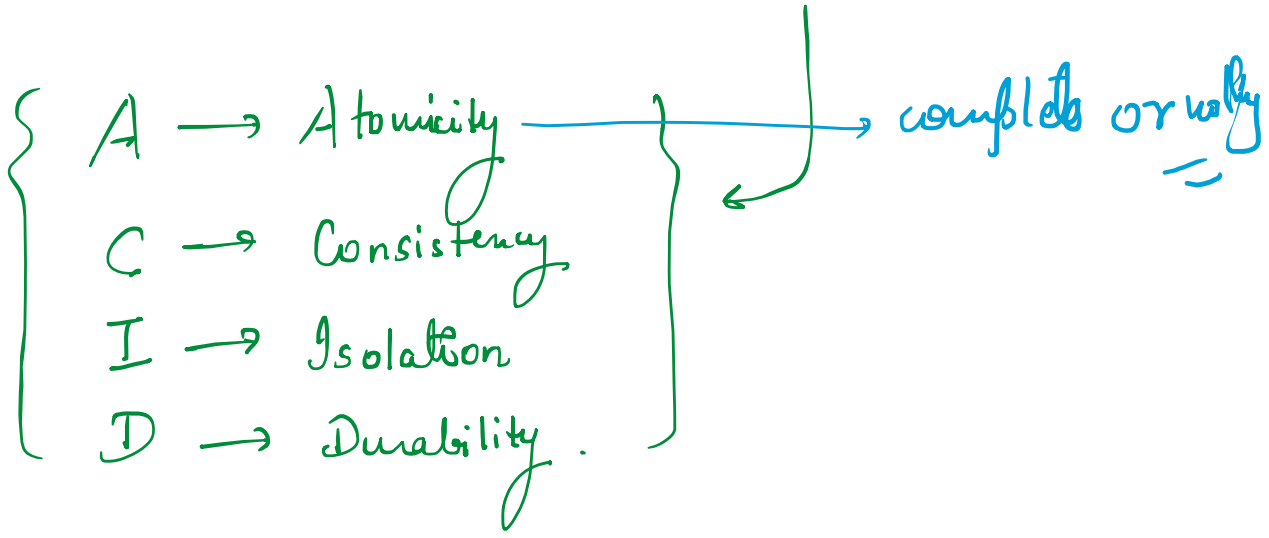


(X) ACID properties

↳ Properties followed by RDBMSs.

↳ Most DB provides

These are some guarantees that DB provides



(*) D: Durability → relative.

Once DB ack's that I've stored data, it persists it forever (ideally)

? How DB ensures once a txn. is committed, it persists?

↳ they store data in 2° mem. — HD, SSD, - where data persists even when power switchedoff.

↳ WALs

Transactions (in DB)

Update students set C -id = 102
where ---

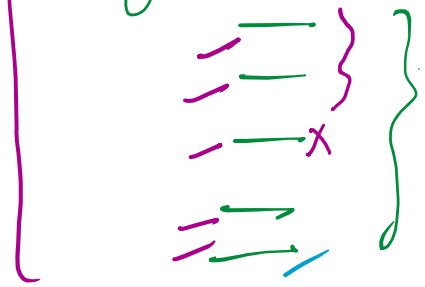
Count (*) students where C -id = 102;

1:15 pm

↓
can give unexpected result if update fails partially or completely.

↙ Soln.

Begin Transaction



all requires.

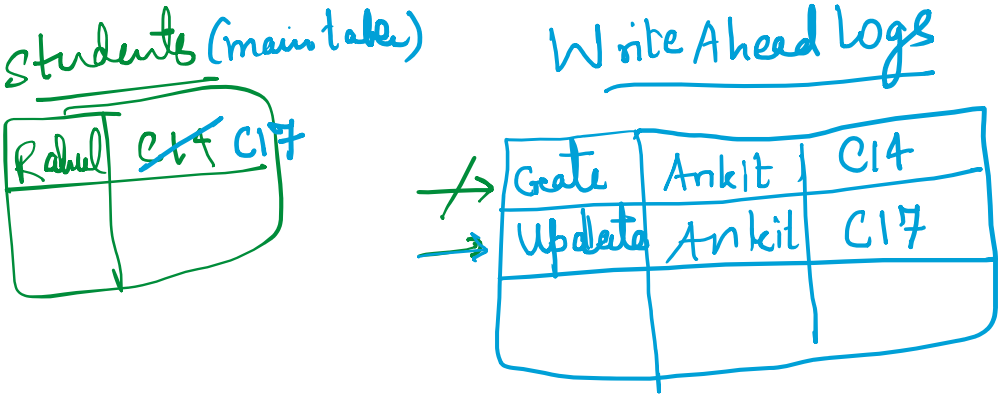
End

⭐ Write Ahead Logs (WALs)

↳ This is a table in DB.

↳ This is a table in DB.
↳ Its an "append-only" → no updates.
↳ Used in DB (CUD operations)

Lets say, I have a student & his cohort is C14. & lets says we want to update it to "C17"
⇒ this student is in "students" main-table



For update (when sql command executed)
First, an entry is made to WAL.
Thereafter, its executed in the main table
↳ stored in DB Main File

∴ WAL
just logs the operations (performed in DB) in sequential manner

Now, there is something called 'CHECKPOINT' in DB-WAL

signifies: all entries in WAL before it got executed

Adv: Writing to WAL is cheaper.

How update on main table is actually executed.?

Ex:

```
Update Students
Set name = 'Rahul'
Where std_id = 102;
```

```
for ( Row row : students){
    if (row[name] == 'Rahul'){
```

```
if (row[name] == "Rahul") {
    row[std_id] = to2;
}
}
```

[First finding then updating] → costly.

If DB goes down admist of this operation is lost forever

Soln

WAL

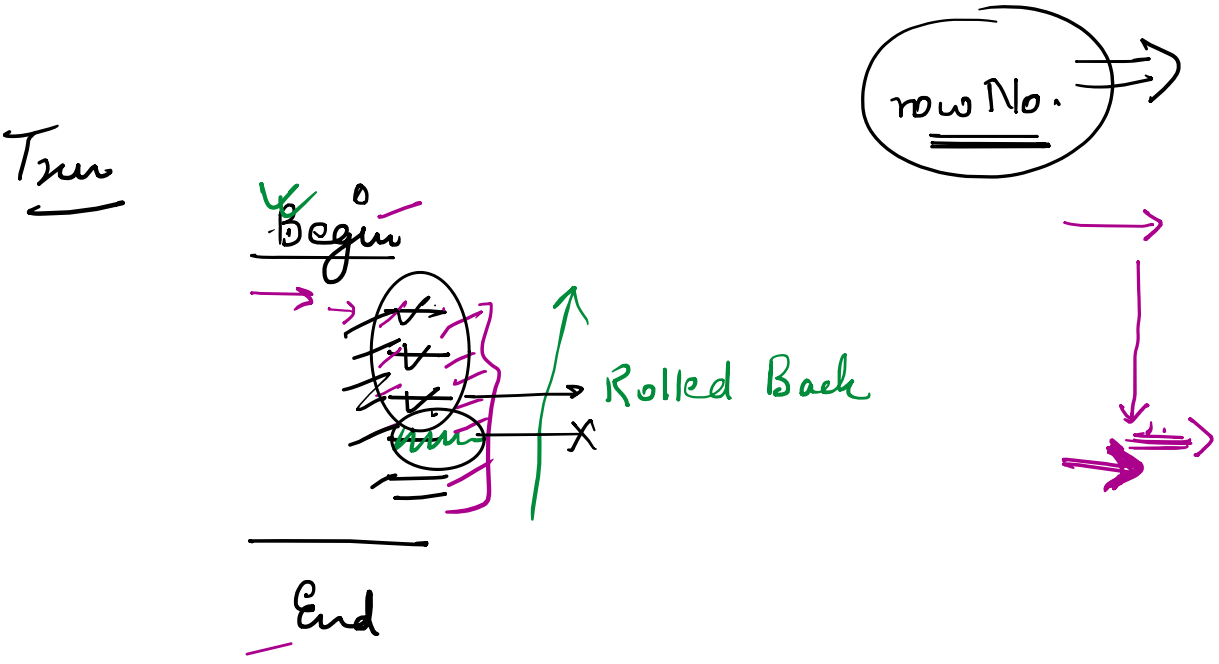
↓

Append-only

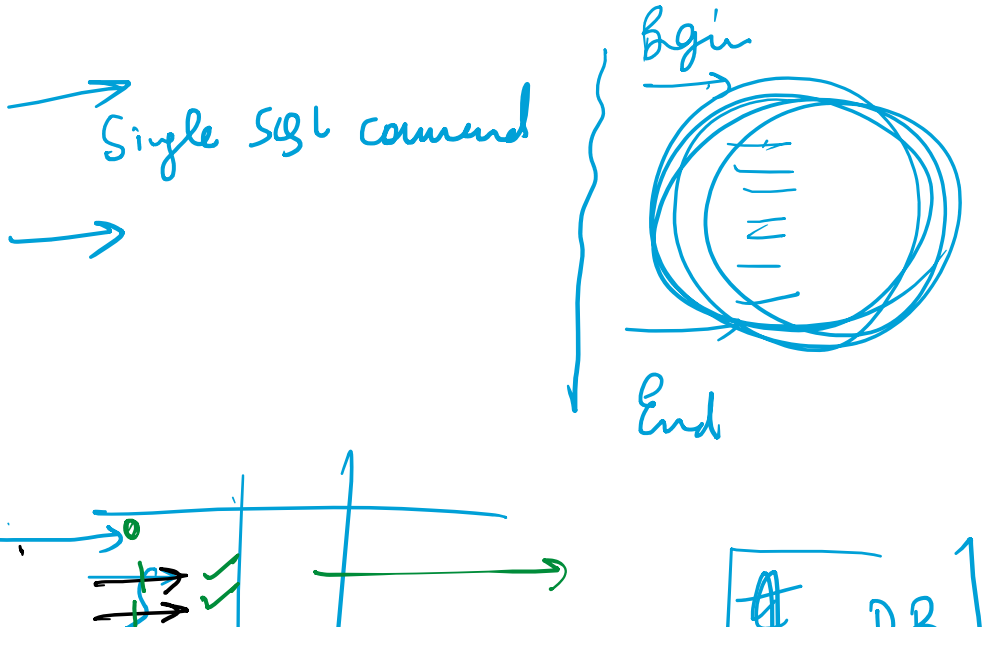
helps in DR (data-recovery)

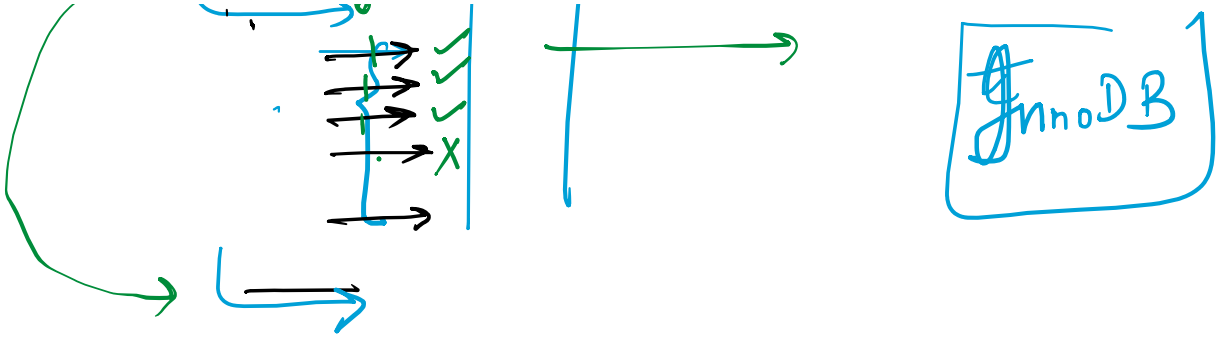
not 100% guaranteed.

Note:- WAL gets cleaned-up periodically.



How RollBacks happen in DB ?




X
InnoDB